

Build a Salmon Redd

Target grade: Grades 3-5

Time: 45 minutes - 1 hour

Team size: 3-4 students

Introduction

Salmon eggs survive best when they are protected within loose gravel that allows oxygen-rich water to flow around them. Too much fine sediment can fill spaces between rocks and make spawning habitats less suitable.

In this lesson, students will investigate what makes a healthy salmon spawning habitat by designing and testing their own model salmon redd, the gravel nest where salmon lay their eggs. Each team will receive a shallow tray, gravel, pebbles, sand, and small beads to represent salmon eggs. Students will then arrange the materials to create a redd that protects the eggs while still allowing water to move through the streambed. To test, teams will pour an equal amount of water through each model stream and observe whether the eggs remain protected and how water travels through the gravel.

Learning Objectives

Students will:

- Explain what a salmon redd is and why salmon build redds in stream gravel
- Identify important conditions salmon eggs need to survive, including protection, flowing water, and oxygen
- Design and test a model salmon redd using different materials
- Observe how different materials affect egg protection
- Connect erosion and sediment upstream to changes in salmon habitat downstream

Materials (Per team):

<u>Material</u>	<u>Quantity</u>
Shallow tray, plastic container, or aluminum pan	1
Gravel	About 1 cup
Small pebbles or rocks	About ½ cup
Sand	About ½ cup
Small beads, dried beans or other to represent salmon eggs	5
Fine soil	1
Water	About 3-4 cups

Small cup or spoon	1
Student observation sheet	1 per student or team

Instructions

1. Begin by introducing the salmon life cycle and explaining how adult salmon return to freshwater streams to lay eggs
2. Explain what a salmon redd is and how female salmon create them
3. Ask students why salmon might bury their eggs instead of leaving them on top of the streambed
 - a. Protection from predators, moving water, other disturbances
4. Introduce the challenge with the following objectives:
 - a. Keep the salmon eggs
 - b. Keep the eggs washing away
 - c. Leave enough open space for water to move through the streambed
5. Give each team their materials (a tray, gravel, pebbles, sand, and salmon eggs) and have students discuss how to arrange their materials before building
 - a. They do not need to use every material
 - b. All 5 eggs must be placed inside the redd
 - c. Eggs cannot be taped, glued, or attached to the tray
6. Give teams around 10-15 minutes to build their salmon redd and ask these guiding questions
 - a. What materials do you think will protect the eggs the best?
 - b. Where do you see spaces where water could move?
 - c. How can you protect the eggs without packing the streambed too tightly?
7. Once all teams finish, conduct a stream flow test
 - a. Give each student the same amount of water
 - b. Slightly pour the water from one end of the tray to represent stream flow
 - c. Have students observe what happens to their streambed and eggs
 - d. Have students record
 - i. How many eggs remained protected and how many eggs moved away from the redd
8. Have students conduct a second test, the **Sediment Storm Test**
 - a. Give each team an equal amount of fine soil or sand and place it at the upstream end of the tray
 - b. Pour another equal amount of water through each model stream
 - c. Have students observe and record
 - i. Where and how the sediment travels
 - ii. How many eggs remained protected and how many eggs moved away from the redd
9. Have a class discussion using some of the following questions:
 - a. Which materials worked best for protecting salmon?

- b. What happened when water flowed through your redd?
 - c. How did the sediment move during the Sediment Storm Test? Where did it collect?
 - d. Why might tightly packed mud or sand be a poor place for salmon eggs?
 - e. If you could rebuild your redd, what would you change?
 - f. What could people do to reduce the amount of excess sediment entering salmon streams?
10. Finish with a final takeaway for what we can do as a community to better protect salmon and stream habitats

